



American International University- Bangladesh (AIUB)
Faculty of Engineering (EEE)

Marks
obtained:

Course Name:	Digital Logic and Circuits	Course Code:	EEE 3101
Semester:	Fall 24-25	Sec:	
Faculty:		Term:	Final
Marks:	20	Assignment Name:	Mid Term Assignment

CO Number	CO1	POI Number	P.a.2.C3	Knowledge Profile:	K2
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Student Name:		Student ID:	
Submission Date:		Department:	

CO	COs/CLOs Statements	K	P	A	Assessed Program Outcome Indicator	BNQF Indicator	Assessment Strategy
CO2	Apply laws of Boolean Algebra for implementing combinational digital logic circuits (gate and transistor level) and for the calculation of performance parameters (Fan-out, Fan-in, Power dissipation, Speed power products, Noise margin) along with Adders, Magnitude Comparators, Encoder, Decoder, Multiplexer and Demultiplexer with the familiarity of issues.	2			P.a.2.C3	FS.2	Assignment (Mid Terms)

SS

- A) Grape View Farm grows a particular type of fruit. Buyer contract says that the customer will take only those which are more than or equal to 7 grams. Now, the farm owner needs a light indication system which can glow different colors of lights to show if the fruit is equal, grater or larger than that stone kept in one of the weight machines. The workers will place a 7-gram stone on one weight machine and the fruit on the other. Red light will glow if less, Yellow for greater and green for exact equal. Design the system applying basic gates. Consider the scales can provide binary output corresponding to its weight and fed to the inputs of your circuit. [10]
- B) Toll Plaza of a bridge has 4 terminals. The central room has one indicator light for the supervisor to see how busy the traffic is. The light glows only when there is two or more than two terminals are collecting tolls. Prepare the function and implement applying an 8 X 1 MUX. [10]

Marking Rubrics:

CP	Assessment Criteria	Evaluation Criteria				Marks
		Poor [1-2]	Average [3-4]	Good [5-6]	Excellent [7-7.5]	
P1	Outline of the standard procedure of digital system design	More than three steps are incorrect or missing and not in correct sequence	One or Two steps of the standard procedure is missing with a one or two steps not in sequence.	All the steps of the procedure have been identified with one or two steps not in correct sequence	All the steps of the procedure have been identified and in correct sequence	
	Digital Triggering Circuit Design.	Design flow has major errors and transistor level design has major flaws.	Design Flow has major error with error carried forward to transistor level design	Design Flow has minor error with error carried forward to transistor level design	Accurate Design Flow with transistor level design having no or minor errors	
P2, P6	Alarm/ Buzzer Design	Alarm design has major flaws which does not comply with the conflicting requirements with major calculation errors.	Alarm design has major flaws which does not comply with the conflicting requirements but with minor calculation errors.	Alarm design is correct and complies to the conflicting requirements but, with major calculation errors.	The alarm design is correct and comply to the requirements with no or minor calculation errors.	
	Limitation	Most of the limitations are irrelevant	Some of the limitations have been addressed	Addressed most of the major limitations	All are correct	
Total Marks Obtained						